

Charlotte Chepngeno Sigei

DATA SCIENTIST | QUANTITATIVE RISK ANALYST | ACTUARIAL PROFESSIONAL

Mbour, Senegal +221 771 547 543 | +254757246589 charlotte.c.sigei@aims-senegal.org
[linkedin.com/in/charlotte-sigey-aa377a254](https://www.linkedin.com/in/charlotte-sigey-aa377a254) github.com/Ccspoet

PROFESSIONAL SUMMARY

AIMS Master's Scholar and Actuarial Science graduate specializing in the intersection of Quantitative Risk Modeling and Social Impact. Expert in applying statistical learning theory and machine learning to financial and development challenges. Proven experience in driving business growth for agri-tech startups and auditing national financial models.

TECHNICAL SKILLS

Data Science: Python (Scikit-learn, Pandas, NumPy), R, SQL, Statistical Learning.
Quantitative: Machine Learning, Stochastic Processes, Financial Modeling, Actuarial Risk.
Business: Startup Growth, Partnership Development, Pitch Deck Creation, Market Research.
Languages: English, Swahili, French (Professional Working Proficiency).

EDUCATION

Master of Science in Data Science 2025 – Present
African Institute for Mathematical Sciences (AIMS), Senegal
— Awarded full academic scholarship for excellence in mathematical sciences.
— Focus: Machine Learning, Stochastic Processes, Optimization, and Reproducible Research.

Bachelor of Science in Actuarial Science 2019 – 2023
Kisii University, Kenya
— Project: Mortality Grading Using Frailty Models – Applied advanced statistical approaches.

PROFESSIONAL EXPERIENCE

Risk and Investment Assistant May 2025 – Sept 2025
Affordable Housing Board (AHB), Kenya
— Financial Modeling: Audited models driving housing finance decisions for the National Program.
— Risk Analysis: Conducted sensitivity analysis on Tenant Purchase Schemes (TPS) structures.
— Data Governance: Enforced strict data protection protocols for sensitive beneficiary data.

Junior Business Development & Strategic Partnerships 2024 – 2025
AgriSync Lab / Farmers Assistant, Kenya
— User Growth: Scaled AI-powered advisory platform base to over 450+ smallholder farmers.
— Strategic Planning: Co-developed investment pitch decks for seed funding and partnerships.
— Market Analysis: Conducted research to identify gaps in agricultural extension services.

CERTIFICATIONS & AWARDS

NVIDIA: Deep Learning Certification | TASS Member | Google: Digital Skills for Africa

References available on request